

Assignment 2 Solutions

Q1 (10 points)

a) Let X denote the number of heads in three tosses of a fair coin.

$$P(X = k) = \binom{3}{k} \left(\frac{1}{2}\right)^3, \quad k = 0, 1, 2, 3.$$

b) PMF as follows:

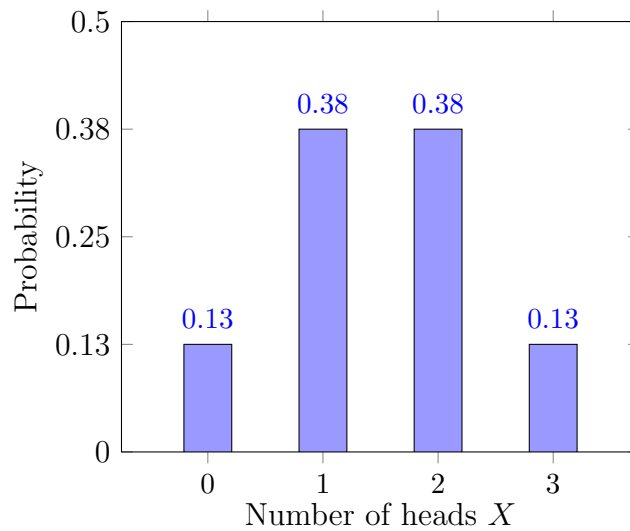


Figure 1: PMF of X = number of heads in 3 fair coin tosses.

c)

$$P(X = 2) = \binom{3}{2} \left(\frac{1}{2}\right)^3 = 3/8.$$

$$P(X \geq 1) = 1 - P(X = 0) = 1 - \frac{1}{8} = \frac{7}{8}.$$

Q2 (10 points)

a) Let $p_X(k) = c(2k + 1)$ for $k = 0, 1, 2, 3$.

$$\begin{aligned}\sum p_X(k) &= 1 \\ 1 &= c[(1) + (3) + (5) + (7)] = 16c \Rightarrow c = \frac{1}{16}.\end{aligned}$$

b)

$$P(X > 1) = c[(5) + (7)] = \frac{1}{16}(12) = \frac{3}{4}.$$

c)

$$\begin{aligned}E[X] &= \sum_x x \cdot p_X(x) \\ &= \frac{1}{16}(2(1) + 1) + 2 \times \frac{1}{16}(2(2) + 1) + 3 \times \frac{1}{16}(2(3) + 1) \\ &= \frac{1}{16}(3) + \frac{1}{16}(10) + \frac{1}{16}(21) \\ &= \frac{1}{16}(34) = \frac{17}{8} = 2.125\end{aligned}$$

d)

$$\begin{aligned}E[X^2] &= \sum_x x^2 p_X(x) \\ &= \frac{1}{16}(2(1) + 1) + 2^2 \times \frac{1}{16}(2(2) + 1) + 3^2 \times \frac{1}{16}(2(3) + 1) \\ &= \frac{3}{16} + \frac{20}{16} + \frac{63}{16} \\ &= \frac{86}{16} = \frac{43}{8} = 5.375\end{aligned}$$

$$\begin{aligned}Var(X) &= E[X^2] - (E[X])^2 \\ &= 5.375 - (2.125)^2 \approx 0.859\end{aligned}$$

Q3 (10 points)

- a) Assuming it takes k coin tosses to obtain the first heads, meaning that the first, $k - 1$ tosses show up as tails. Note that the random variable, X can take on infinitely many values and that the probability of a tails showing up is $1 - p$. Hence, pmf of X is given by:

$$p_X(k) = (1 - p)^{k-1}p \quad k = 1, 2, \dots,$$

b)

$$\begin{aligned} E(X) &= \sum_{k=1}^{\infty} kp_X(k) \\ &= \sum_{k=1}^{\infty} k(1 - p)^{k-1}p \\ &= p \sum_{k=1}^{\infty} k(1 - p)^{k-1} \end{aligned}$$

Since:

$$\sum_{k=1}^{\infty} k(1 - p)^{k-1} = -\frac{d}{dp} \left(\sum_{k=0}^{\infty} (1 - p)^k \right)$$

$$E[X] = -p \frac{d}{dp} \left(\sum_{k=0}^{\infty} (1 - p)^k \right)$$

Since:

$$\sum_{k=1}^{\infty} (1 - p)^k = \frac{1 - p}{p}$$

$$\begin{aligned} E[X] &= -p \frac{d}{dp} \left(\frac{1 - p}{p} \right) = p \frac{d}{dp} \left(\frac{p - 1}{p} \right) = p \frac{d}{dp} \left(1 - \frac{1}{p} \right) \\ &= p \left(\frac{1}{p^2} \right) \\ &= \frac{1}{p} \end{aligned}$$

c)

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

$$\begin{aligned} E(X^2) &= \sum_{k=1}^{\infty} k^2 p_X(k) \\ &= \sum_{k=1}^{\infty} k^2 (1-p)^{k-1} p \\ &= p \sum_{k=1}^{\infty} k^2 (1-p)^{k-1} \end{aligned}$$

The summation $\sum_{k=1}^{\infty} k^2 (1-p)^{k-1}$ can be split as:

$$\sum_{k=1}^{\infty} k^2 (1-p)^{k-1} = \sum_{k=1}^{\infty} k(k-1)(1-p)^{k-1} + \sum_{k=1}^{\infty} k(1-p)^{k-1}$$

Hence:

$$E[X] = p \left(\sum_{k=1}^{\infty} k(k-1)(1-p)^{k-1} + \sum_{k=1}^{\infty} k(1-p)^{k-1} \right)$$

Earlier we saw that:

$$\sum_{k=1}^{\infty} k(1-p)^k - 1 = \frac{d}{dp} \left(\sum_{k=0}^{\infty} (1-p)^k \right) = \frac{1}{p^2}$$

Similarly:

$$\begin{aligned} \sum_{k=1}^{\infty} k(k-1)(1-p)^{k-1} &= (1-p) \sum_{k=1}^{\infty} k(k-1)(1-p)^{k-2} \\ &= (1-p) \frac{d^2}{dp^2} \left(\sum_{k=0}^{\infty} (1-p)^k \right) = \frac{2(1-p)}{p^3} \end{aligned}$$

Finally:

$$\begin{aligned} E[X] &= p \left(\sum_{k=1}^{\infty} k(k-1)(1-p)^{k-1} + \sum_{k=1}^{\infty} k(1-p)^{k-1} \right) \\ &= p \left(\frac{2(1-p)}{p^3} + \frac{1}{p^2} \right) = p \left(\frac{2-2p}{p^3} + \frac{p}{p^3} \right) = p \left(\frac{2-p}{p^3} \right) \\ &= \frac{2-p}{p^2} \end{aligned}$$

Hence:

$$\begin{aligned} \text{Var}(x) &= E[X^2] - E[X]^2 = \frac{2-p}{p^2} - \frac{1}{p^2} \\ &= \frac{1-p}{p^2} \end{aligned}$$

For a geometric random variable X with parameter p , the expectation and variance are:

$$E[X] = \frac{1}{p}, \quad \text{Var}(X) = \frac{1-p}{p^2}.$$

Q4 (10 points)

- a) Binomial
- b) Uniform
- c) Geometric

Q5 (10 points)

This question is a binomial distribution problem; and is similar to tossing a biased coin 3 times. With the probability of getting a head(a call from a telemarketer). The four possible outcomes are 0 calls, 1 call, 2 calls, and 3 calls. The following are the probabilities for the outcomes:

$$P(X = 0) = {}^3C_0 \left(\frac{7}{8}\right)^3 = \frac{343}{512}$$

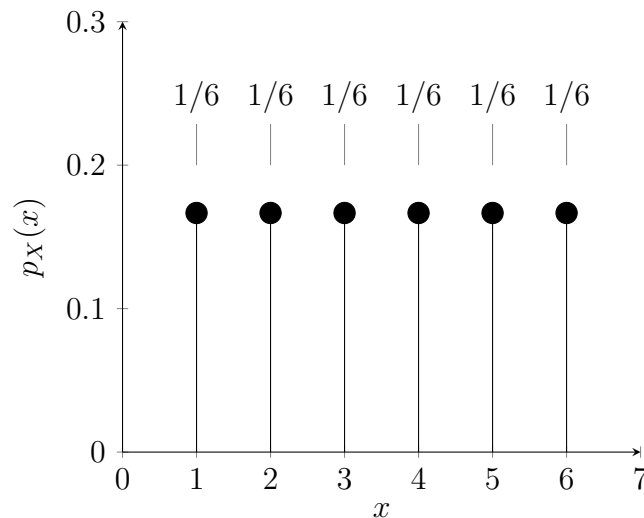
$$P(X = 1) = {}^3C_1 \left(\frac{1}{8}\right) \left(\frac{7}{8}\right)^2 = \frac{147}{512}$$

$$P(X = 2) = {}^3C_2 \left(\frac{1}{8}\right)^2 \left(\frac{7}{8}\right) = \frac{21}{512}$$

$$P(X = 3) = {}^3C_3 \left(\frac{1}{8}\right)^3 = \frac{1}{512}$$

Q6 (10 points)

$p_X(x)$:



a)

Expected value rule:

$$\begin{aligned}
 E[g(x)] &= \sum_x g(x)p_X(x) \\
 E[X^2 + 1] &= \sum_x (x^2 + 1)p_X(x) \\
 &= (1^2 + 1)\frac{1}{6} + (2^2 + 1)\frac{1}{6} + (3^2 + 1)\frac{1}{6} + (4^2 + 1)\frac{1}{6} + (5^2 + 1)\frac{1}{6} + (6^2 + 1)\frac{1}{6} \\
 &= \frac{2}{6} + \frac{5}{6} + \frac{10}{6} + \frac{17}{6} + \frac{26}{6} + \frac{37}{6} \\
 &= \frac{97}{6}
 \end{aligned}$$

b)

For linear functions:

$$\begin{aligned}
 E[X] &= 3.5 \\
 E[2X + 1] &= 2E[X] + 1 \\
 E[2X + 1] &= 2(3.5) + 1 = 8
 \end{aligned}$$

Q7 (10 points)

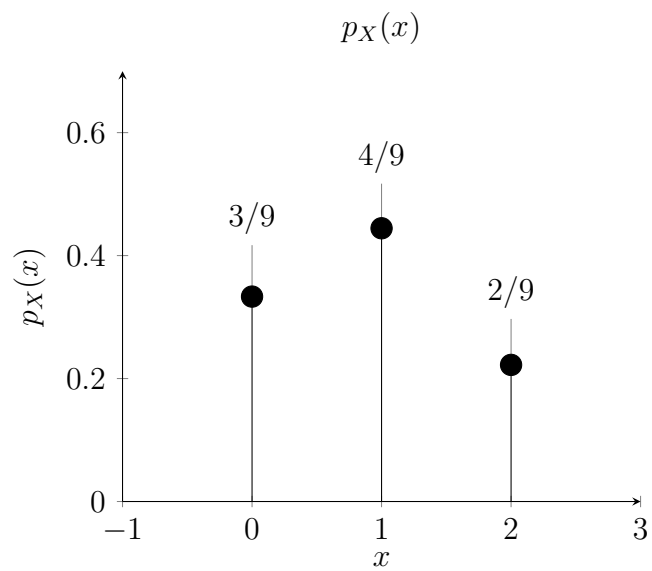
a)

X	Roll 1=1	Roll 1=2	Roll 1=3
Roll 2=1	0	1	2
Roll 2=2	1	0	1
Roll 2=3	2	1	0

Table 1: $X = |\text{Roll 1} - \text{Roll 2}|$

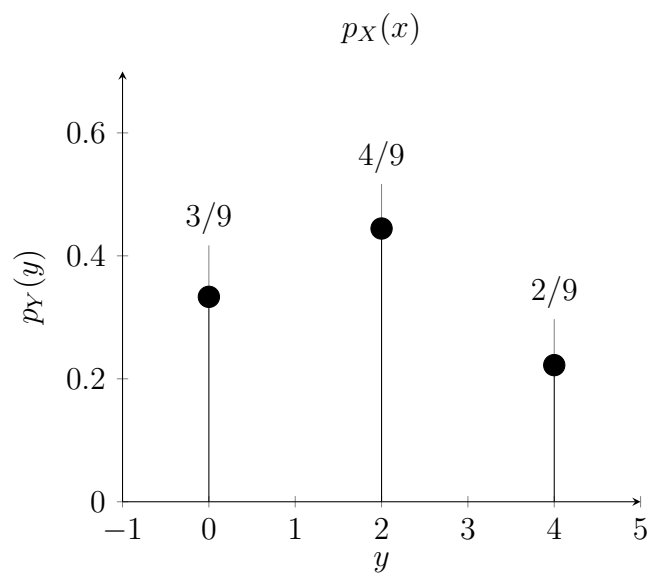
$$X \in \{0, 1, 2\}$$

$$\begin{aligned}
 P(X = 0) &= \frac{3}{9} \\
 P(X = 1) &= \frac{4}{9} \\
 P(X = 2) &= \frac{2}{9}
 \end{aligned}$$



b)

$$Y \in \{0, 2, 4\}$$



c)

$$\begin{aligned}
 \mu_Y = E[Y] &= \sum_y y \cdot p_Y(y) \\
 &= 0 \cdot \frac{3}{9} + 2 \cdot \frac{4}{9} + 4 \cdot \frac{2}{9} \\
 &= \frac{16}{9}
 \end{aligned}$$

d)

$$\text{Var}(Y) = E[Y^2] - (\mu_Y)^2$$

$$\begin{aligned} E[Y^2] &= \sum_y y^2 \cdot p_Y(y) \\ &= 0^2 \cdot \frac{3}{9} + 2^2 \cdot \frac{4}{9} + 4^2 \cdot \frac{2}{9} \\ &= \frac{16}{9} + \frac{32}{9} \\ &= \frac{48}{9} \end{aligned}$$

$$\text{Var}(Y) = \frac{48}{9} - \left(\frac{16}{9}\right)^2 = \frac{176}{81} \approx 2.17$$

Q8 (10 points)

a)

		$X =$					
		1	2	3	4	5	6
$Y =$	0	$\frac{1}{24}$	$\frac{1}{24}$	$\frac{1}{24}$	$\frac{1}{24}$	0	0
	1	$\frac{3}{24}$	$\frac{2}{24}$	$\frac{1}{24}$	0	0	0
	2	0	$\frac{1}{24}$	$\frac{2}{24}$	$\frac{3}{24}$	$\frac{4}{24}$	$\frac{4}{24}$

b)

$$\begin{aligned} p_Y(0) &= \sum_x p_{X|Y}(x, 0) = \frac{4}{24} = \frac{1}{6} \\ p_Y(1) &= \sum_x p_{X|Y}(x, 1) = \frac{6}{24} = \frac{1}{4} \\ p_Y(2) &= \sum_x p_{X|Y}(x, 2) = \frac{14}{24} = \frac{7}{12} \end{aligned}$$

c)

$$E[X] = \sum_x x \cdot p_X(x)$$

$$E[X] = \sum_x x \left(\sum_y p_{X|Y}(x, y) \right)$$

$$E[X] = \frac{4}{24} + 2 \times \frac{4}{24} + 3 \times \frac{4}{24} + 4 \times \frac{4}{24} + 5 \times \frac{4}{24} + 6 \times \frac{4}{24}$$

$$E[X] = 3.5$$

d)

$$p_{X|Y}(x|2) = \begin{cases} 0, & x = 1 \\ \frac{1}{14}, & x = 2 \\ \frac{2}{14}, & x = 3 \\ \frac{3}{14}, & x = 4 \\ \frac{4}{14}, & x = 5 \\ \frac{4}{14}, & x = 6 \end{cases}$$

Q9 (10 points)

a)

Joint PMF of X_1 and X_2 :

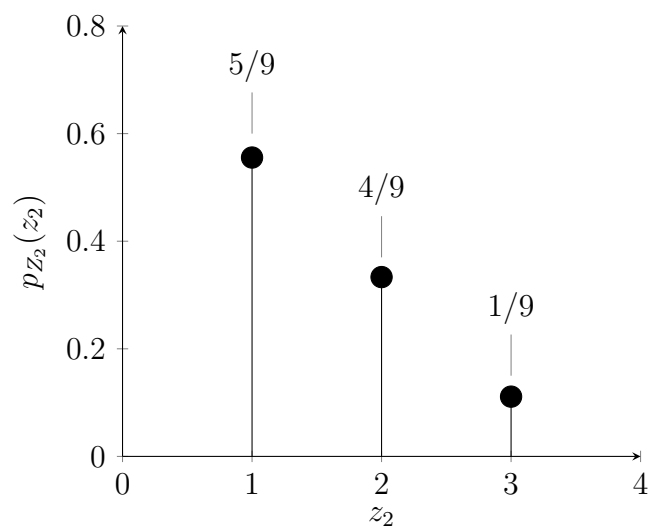
$$P(Z_2 = 1) = p_{X_1|X_2}(1, 1) + p_{X_1|X_2}(1, 2) + p_{X_1|X_2}(1, 3) + p_{X_1|X_2}(2, 1) + p_{X_1|X_2}(3, 1) = \frac{5}{9}$$

$$P(Z_2 = 2) = p_{X_1|X_2}(2, 2) + p_{X_1|X_2}(2, 3) + p_{X_1|X_2}(3, 2) = \frac{3}{9}$$

$$P(Z_2 = 3) = p_{X_1|X_2}(3, 3) = \frac{1}{9}$$

		$X_1 =$		
		1	2	3
$X_2 =$	1	$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$
	2	$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$
	3	$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$

PMF of Z_2 :

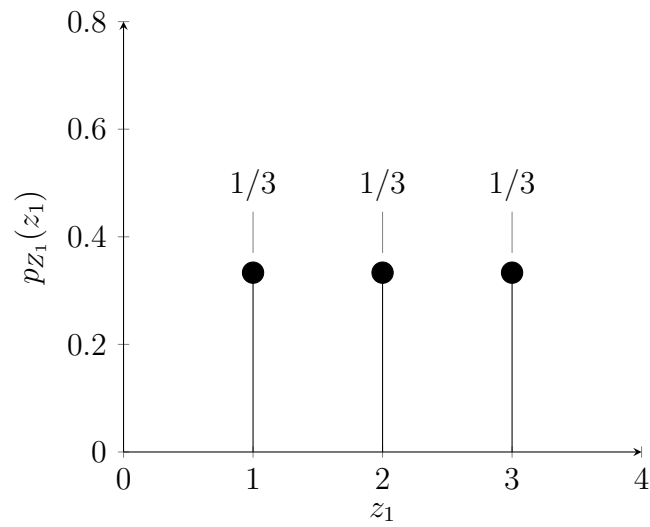


Mean of Z_2 :

$$\begin{aligned}
 E[Z_2] &= 1 \left(\frac{5}{9} \right) + 2 \left(\frac{4}{9} \right) + 3 \left(\frac{1}{9} \right) \\
 &= \frac{5}{9} + \frac{6}{9} + \frac{3}{9} \\
 &= \frac{14}{9}
 \end{aligned}$$

b)

PMF of Z_1



Mean of Z_1 :

$$\begin{aligned} E[Z_1] &= 1 \left(\frac{1}{3} \right) + 2 \left(\frac{1}{3} \right) + 3 \left(\frac{1}{3} \right) \\ &= 2 \end{aligned}$$

Mean of Y :

$$\begin{aligned} Y &= 2Z_1 + Z_2 \\ E[Y] &= 2E[Z_1] + E[Z_2] \\ &= 2(2) + \frac{14}{9} \\ &= \frac{50}{9} \approx 5.56 \end{aligned}$$

Q10 (10 points)

Assuming each roll of the die is independent with probability $\frac{1}{6}$, the distribution is multinomial in terms of X and Y . The probability of obtaining a 1 on any roll is $\frac{1}{6}$, a 2 on any roll is, $\frac{1}{6}$, and neither a 1 nor a 2 on any roll is $\frac{4}{6}$. The joint PMF of both X and Y are given by:

$$\begin{aligned} p_{X,Y}(x,y) &= {}^4C_x {}^{4-x}C_y {}^{4-x-y}C_{4-x-y} \left(\frac{1}{6}\right)^x \left(\frac{1}{6}\right)^y \left(\frac{4}{6}\right)^{4-x-y} \\ p_{X,Y}(x,y) &= \frac{4!}{(4-x)!x!} \frac{(4-x)!}{(4-x-y)!y!} \frac{(4-x-y)!}{(4-x-y)!0!} \left(\frac{1}{6}\right)^{x+y} \left(\frac{4}{6}\right)^{4-x-y} \\ &= \frac{4!}{x!y!(4-x-y)!} \left(\frac{1}{6}\right)^{x+y} \left(\frac{4}{6}\right)^{4-x-y} \end{aligned}$$

Note that:

$$X, Y \in \{0, 1, 2, 3, 4\}, \text{ and } X + Y \leq 4$$

And:

$$p_{X,Y}(x,y) = 0, \text{ } X + Y > 4$$

Finally:

$$p_{X,Y}(x,y) = \begin{cases} \frac{4!}{x!y!(4-x-y)!} \left(\frac{1}{6}\right)^{x+y} \left(\frac{4}{6}\right)^{4-x-y} & X, Y \in \{0, 1, 2, 3, 4\}, \text{ and } X + Y \leq 4 \\ 0 & X + Y > 4 \end{cases}$$

Optional sanity check:

$$\begin{aligned} p_{X,Y}(0,0) &= \frac{4!}{0!0!4!} \left(\frac{1}{6}\right)^0 \left(\frac{4}{6}\right)^4 = \left(\frac{4}{6}\right)^4 = \frac{16}{81} \\ p_{X,Y}(0,1) &= \frac{4!}{0!1!3!} \left(\frac{1}{6}\right)^1 \left(\frac{4}{6}\right)^3 = 4 \left(\frac{1}{6}\right)^1 \left(\frac{4}{6}\right)^3 = \frac{16}{81} \\ p_{X,Y}(0,2) &= \frac{4!}{0!2!2!} \left(\frac{1}{6}\right)^2 \left(\frac{4}{6}\right)^2 = 6 \left(\frac{1}{6}\right)^2 \left(\frac{4}{6}\right)^2 = \frac{2}{27} \\ p_{X,Y}(0,3) &= \frac{4!}{0!3!1!} \left(\frac{1}{6}\right)^3 \left(\frac{4}{6}\right)^1 = 4 \left(\frac{1}{6}\right)^3 \left(\frac{4}{6}\right)^1 = \frac{1}{81} \\ p_{X,Y}(0,4) &= \frac{4!}{0!4!0!} \left(\frac{1}{6}\right)^4 \left(\frac{4}{6}\right)^0 = \left(\frac{1}{6}\right)^4 = \frac{1}{1296} \end{aligned}$$

$$\begin{aligned}
p_{X,Y}(1,0) &= \frac{4!}{1!0!3!} \left(\frac{1}{6}\right)^1 \left(\frac{4}{6}\right)^3 = 4 \left(\frac{1}{6}\right)^1 \left(\frac{4}{6}\right)^3 = \frac{16}{81} \\
p_{X,Y}(1,1) &= \frac{4!}{1!1!2!} \left(\frac{1}{6}\right)^2 \left(\frac{4}{6}\right)^2 = 12 \left(\frac{1}{6}\right)^2 \left(\frac{4}{6}\right)^2 = \frac{4}{27} \\
p_{X,Y}(1,2) &= \frac{4!}{1!2!1!} \left(\frac{1}{6}\right)^3 \left(\frac{4}{6}\right)^1 = 12 \left(\frac{1}{6}\right)^3 \left(\frac{4}{6}\right)^1 = \frac{1}{27} \\
p_{X,Y}(1,3) &= \frac{4!}{1!3!0!} \left(\frac{1}{6}\right)^4 \left(\frac{4}{6}\right)^0 = 4 \left(\frac{1}{6}\right)^4 = \frac{1}{324} \\
p_{X,Y}(1,4) &= 0 \\
p_{X,Y}(2,0) &= \frac{4!}{2!0!2!} \left(\frac{1}{6}\right)^2 \left(\frac{4}{6}\right)^2 = 6 \left(\frac{1}{6}\right)^2 \left(\frac{4}{6}\right)^2 = \frac{2}{27} \\
p_{X,Y}(2,1) &= \frac{4!}{2!1!1!} \left(\frac{1}{6}\right)^3 \left(\frac{4}{6}\right)^1 = 12 \left(\frac{1}{6}\right)^3 \left(\frac{4}{6}\right)^1 = \frac{1}{27} \\
p_{X,Y}(2,2) &= \frac{4!}{2!2!0!} \left(\frac{1}{6}\right)^4 \left(\frac{4}{6}\right)^0 = 6 \left(\frac{1}{6}\right)^4 = \frac{1}{216} \\
p_{X,Y}(2,3) &= 0 \\
p_{X,Y}(2,4) &= 0 \\
p_{X,Y}(3,0) &= \frac{4!}{3!0!1!} \left(\frac{1}{6}\right)^3 \left(\frac{4}{6}\right)^1 = 4 \left(\frac{1}{6}\right)^3 \left(\frac{4}{6}\right)^1 = \frac{1}{81} \\
p_{X,Y}(3,1) &= \frac{4!}{3!1!0!} \left(\frac{1}{6}\right)^4 \left(\frac{4}{6}\right)^0 = 4 \left(\frac{1}{6}\right)^4 = \frac{1}{324} \\
p_{X,Y}(3,2) &= 0 \\
p_{X,Y}(3,3) &= 0 \\
p_{X,Y}(3,4) &= 0 \\
p_{X,Y}(4,0) &= \frac{4!}{4!0!0!} \left(\frac{1}{6}\right)^4 \left(\frac{4}{6}\right)^0 = \left(\frac{1}{6}\right)^4 = \frac{1}{1296} \\
p_{X,Y}(4,1) &= 0 \\
p_{X,Y}(4,2) &= 0 \\
p_{X,Y}(4,3) &= 0 \\
p_{X,Y}(4,4) &= 0
\end{aligned}$$

$p_{X,Y}(x,y)$		$X =$					
		0	1	2	3	4	$p_Y(x,y)$
$Y =$	0	$\frac{16}{81}$	$\frac{16}{81}$	$\frac{2}{27}$	$\frac{1}{81}$	$\frac{1}{1296}$	$\frac{625}{1296}$
	1	$\frac{16}{81}$	$\frac{4}{27}$	$\frac{1}{27}$	$\frac{1}{324}$	0	$\frac{125}{324}$
	2	$\frac{2}{27}$	$\frac{1}{27}$	$\frac{1}{216}$	0	0	$\frac{25}{216}$
	3	$\frac{1}{81}$	$\frac{1}{324}$	0	0	0	$\frac{5}{324}$
	4	$\frac{1}{1296}$	0	0	0	0	$\frac{1}{1296}$
	$p_X(x,y)$	$\frac{625}{1296}$	$\frac{125}{324}$	$\frac{25}{216}$	$\frac{5}{324}$	$\frac{1}{1296}$	

Q11 (10 points)

Let X be the number of heads in three fair coin tosses. The unconditional PMF of X is given by:

$$P(X = k) = {}^3C_k \left(\frac{1}{2}\right)^3, \quad k \in \{0, 1, 2, 3\}$$

so:

$$\begin{aligned} P(X = 0) &= \frac{1}{8} \\ P(X = 1) &= \frac{3}{8} \\ P(X = 2) &= \frac{3}{8} \\ P(X = 3) &= \frac{1}{8} \end{aligned}$$

The event A is $\{X \geq 1\}$; its probability is

$$\begin{aligned} P(A) &= 1 - P(X = 0) \\ &= 1 - \frac{1}{8} \\ &= \frac{7}{8} \end{aligned}$$

(a) Conditional PMF.

For $x \in \{1, 2, 3\}$

$$\begin{aligned} p_{X|A}(x) &= P(X = x \mid A) = \frac{P(X = x)}{P(A)} \\ &= \frac{P(X = x)}{7/8} \\ &= \frac{8}{7}P(X = x) \end{aligned}$$

Thus:

$$\begin{aligned} P(X = 1 \mid A) &= \frac{3/8}{7/8} = \frac{3}{7} \\ P(X = 2 \mid A) &= \frac{3/8}{7/8} = \frac{3}{7} \\ P(X = 3 \mid A) &= \frac{1/8}{7/8} = \frac{1}{7} \end{aligned}$$

And $P(X = 0 \mid A) = 0$

Optional Check: $3/7 + 3/7 + 1/7 = 1$

(b) Conditional expectation.

Compute

$$\begin{aligned} E[X \mid A] &= \sum_{x=1}^3 x P(X = x \mid A) \\ &= 1 \cdot \frac{3}{7} + 2 \cdot \frac{3}{7} + 3 \cdot \frac{1}{7} \\ &= \frac{3 + 6 + 3}{7} = \frac{12}{7} \approx 1.71 \end{aligned}$$

Q12 (10 points)

By the interpretation of the mean as the average of a large number of selected experiments, the average CGPA should be close to $E[Y]$.

By the total expectation theorem:

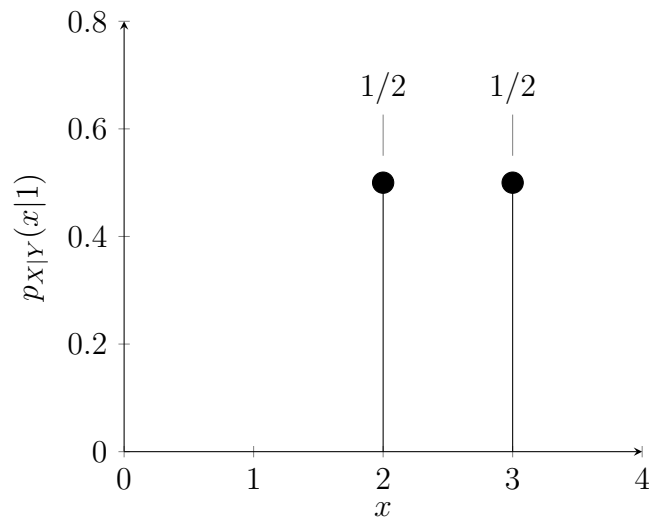
$$\begin{aligned} E[Y] &= p_X(0)E[Y|X=0] + p_X(1)E[Y|X=1] \\ &= 0.4(3.2) + 0.6(3) \\ &= 3.08 \end{aligned}$$

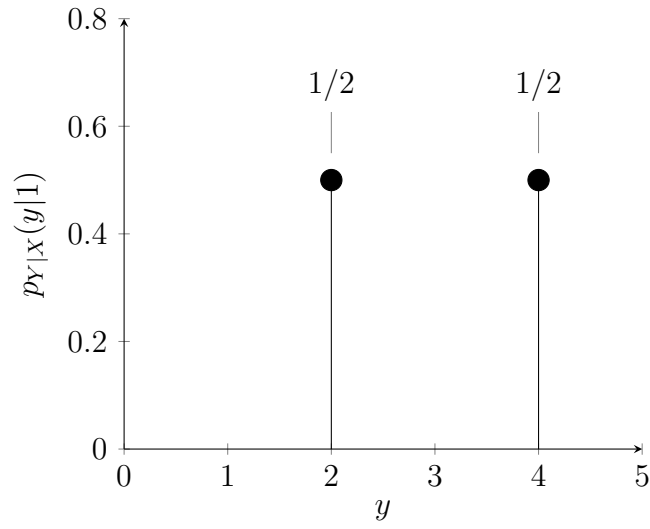
Q13 (10 points)

a)

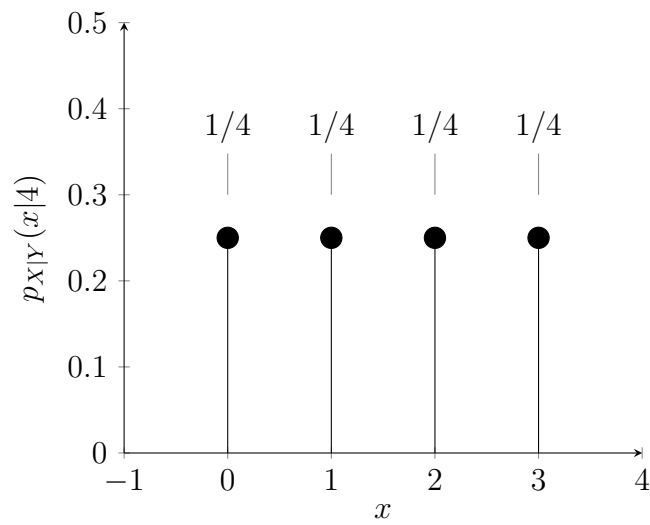
No.

b)





c)



$$\begin{aligned}
 E[X|Y=4] &= \sum_x x \cdot p_{X|Y}(x|4) \\
 &= 0 \left(\frac{1}{4} \right) + 1 \left(\frac{1}{4} \right) + 2 \left(\frac{1}{4} \right) + 3 \left(\frac{1}{4} \right) \\
 &= \frac{3}{2} = 1.5
 \end{aligned}$$

d)

From $p_{X|Y}(X|Y)$:

$$p_R(1) = 0$$

$$p_R(2) = \frac{3}{8}$$

$$p_R(3) = \frac{1}{8}$$

$$p_R(4) = \frac{4}{8}$$

Hence:

$$\begin{aligned}\mu_R &= 2 \left(\frac{3}{8} \right) + 3 \left(\frac{1}{8} \right) + 4 \left(\frac{4}{8} \right) \\ &= \frac{25}{8} = 3.125\end{aligned}$$

Q14 (10 points)

a)

$$p_X(x) = \sum_y p_{X,Y}(x, y)$$

$$p_X(2) = \sum_y p_{X,Y}(2, y)$$

$$p_X(2) = p_{X,Y}(2, 2) + p_{X,Y}(2, 4) + p_{X,Y}(2, 6)$$

$$p_X(2) = \frac{1}{12} + \frac{2}{12} + \frac{1}{12}$$

$$p_X(2) = \frac{4}{12} = \frac{1}{3}$$

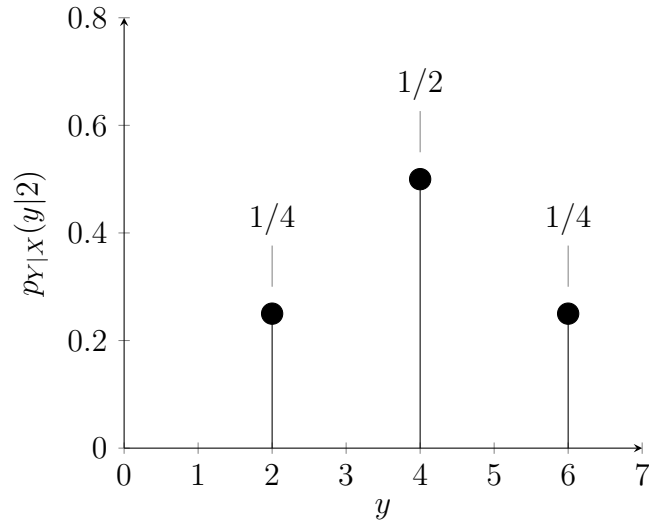
b)

$$p_{Y|X}(y|x) = \frac{p_{X,Y}(x, y)}{p_X(x)}$$

$$p_{Y|X}(2|2) = \frac{p_{X,Y}(2, 2)}{p_X(2)} = \frac{\frac{1}{12}}{\frac{1}{3}} = \frac{1}{4}$$

$$p_{Y|X}(4|2) = \frac{p_{X,Y}(2, 4)}{p_X(2)} = \frac{\frac{2}{12}}{\frac{1}{3}} = \frac{1}{2}$$

$$p_{Y|X}(6|2) = \frac{p_{X,Y}(2,6)}{p_X(2)} = \frac{\frac{1}{12}}{\frac{1}{3}} = \frac{1}{4}$$



$$\begin{aligned} E[Y|X=2] &= 2 \left(\frac{1}{4} \right) + 4 \left(\frac{1}{2} \right) + 6 \left(\frac{1}{4} \right) \\ &= \frac{1}{2} + 2 + \frac{3}{2} \\ &= 4 \end{aligned}$$

d)

$$a = \frac{2}{12}, b = \frac{1}{12}, \text{ and } c = \frac{1}{12}$$

With these choices, all the rows have the same relative likelihood among the members of the row. This implies that having information about y does not change the relative likelihoods of x 's. This implies independence. A similar argument can be applied to the columns as well.